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TYPES AND CLASSIFICATION SYSTEMS OF VENOUS REFLUX IN CHRONIC VENOUS DISEASE: A NARRATIVE LITERATURE SYNTHESIS

TYPES AND CLASSIFICATION SYSTEMS OF VENOUS REFLUX IN CHRONIC VENOUS DISEASE: NARRATIVE SYNTHESIS OF LITERATURE

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Summary

Ob Objectives. Chronic venous disease currently represents one of the most common causes of hospitalization in surgical wards, due to its high incidence and a wide spectrum of clinical manifestations. The multitude of hemodynamic variants of venous reflux that determine the clinical picture and trophic changes in patients with chronic venous disease also explains the existence of various surgical techniques aimed at correcting hemodynamic disorders based on the type of pathological reflux, rather than necessarily relying on the definitive abolition of the saphenous trunk. Special attention is given to surgical interventions that preserve saphenous veins, which have demonstrated advantages for a certain category of patients. The aim of this work is to conduct a narrative review of the types and classification systems of venous reflux in chronic venous disease and their role in selecting the surgical treatment strategy.

Materials and Methods. A structured search was conducted in the electronic databases Google Scholar, PubMed, HINARI, and ResearchGate, considering relevant articles that describe hemodynamic disorders in chronic venous disease published in the last 13 years. The search terms (in English) used were: "varicose veins," "chronic venous insufficiency," "hemodynamic disorders in chronic venous insufficiency," "types of hemodynamic disorders," "chronic venous disease - quality of life," "techniques of treatment for varicose veins," "preservation of saphenous veins," "CHIVA," "ASVAL."

Conclusions. Based on the type of hemodynamic disorders and the clinical profile of the patient, it becomes possible to select an appropriate surgical strategy tailored to each case — with or without preservation of the saphenous trunk — which would ensure long-term therapeutic efficiency for patients with chronic venous disease.

Keywords: chronic venous disease, hemodynamic disorders, reflux, preservation of saphenous veins

Summary

Objectives. Chronic venous disease is currently one of the most common causes of hospitalizations in surgical departments due to its high incidence and a wide range of clinical manifestations. The multitude of hemodynamic variations of venous reflux determines t he clinical picture and trophic changes in patients with chronic venous disease, which also explains the existence of different surgical techniques aimed at correcting hemodynamic disorders depending on the type of pathological reflux; this is not always based on the complete destruction of the saphenous trunk. A significant focus is on interventions that preserve the saphenous trunk, which have shown their advantages in a specific category of patients. Thus, we conducted a narrative review of studies aimed at describing the types and classification systems of venous reflux in chronic venous disease and their role in the selection of surgical treatment.

Material and methods. A structured search was performed in the Google Scholar, PubMed, HINARI and ResearchGate electronic databases, taking into account the relevant articles, which refer to the description of hemodynamic disorders in chronic venous disease, published in the last 13 years. The search terms (in English) used were: "varicose veins", "chronic venous insufficiency", " hemodynamic disorders in chronic venous insufficiency", "types of hemodynamic disorders", "chronic venous disease - quality of life", "techniques of treatment for varicose veins", "preservation of saphenous veins", "CHIVA", "ASVAL".

Conclusions. Based on the type of hemodynamic disorders and the patient's clinical profile, it becomes possible to select a suitable surgical strategy, specific to each individual case - with or without preservation of the saphenous trunk, which will ensure long-term therapeutic efficacy in patients with chronic venous disease.

Keywords: chronic venous disease, hemodynamic disorders, reflux, preservation of saphenous veins

Introduction

Chronic venous disease (CVI) of the lower limbs is one of the most widespread vascular pathologies worldwide [1]. Multiple scientific studies in the field have It has been demonstrated that CVI represents a current public health issue due to its high prevalence, high costs, and negative impact on quality of life [2]. The estimated prevalence of varicose veins in the lower limbs ranges from 5-30%.

in the adult population, with a higher frequency in women (3:1). Despite the fact that most patients present t he disease in its early stages, approximately 10% of them are in advanced stages at the time of presentation [3]. According to the international survey *Vein Consult Program* (VCP), conducted by the International Union of Phlebology in 20 countries, the global incidence of CVI was 83.6%. Additionally, the importance of adequate screening for patients with CVI has been demonstrated, especially for those with low severity [4]. Furthermore, there is evidence that CVI remains, to date, an underdiagnosed and undertreated pathology [5].

The etiopathogenesis of CVI is based on both genetic factors and prolonged exposure to extrinsic risk factors, which together lead to significant biophysical and bioche mical changes in the venous system and generate a complex vascular response [6]. In the context of the CEAP (Clinical, Etiological, Anatomical, Physiopathological) classification, the spectrum of CVI manifestations is extremely variable - ranging from asymptomatic disease with only cosmetic manifestations to severe symptoms. The clinical picture may include telangiectasias, reticular veins, varicose veins, edema, pigmentation, eczema, lipodermatosclerosis, white atrophy, and venous ulceration [7]. At the same time, assessing the anatomical extent of pathological venous reflux through Doppler Duplex scanning is necessary, as well as determining the etiology and physiopathology of varicose veins [8]. Although the Doppler Duplex examination represents the gold standard for diagnosing CVI and determining the affected venous system – superficial, deep, or perforating, additional investigations may sometimes be necessary to establish a definitive diagnosis, especially if the deep venous system is involved and in cases of recurrent disease after surgical treatment [2, 9].

Traditionally, the surgical treatment of CVI involves the removal of the incompetent saphenous trunk, either through open surgery or, increasingly often, through endovenous methods. However, according to current views, minimally invasive treatment modalities that preserve the saphenous trunk, such as CHIVA (*A mbulatory Conservative Hemodynamic Treatment of Venous Insufficiency*) and ASVAL (*Ambulatory Selective Varices Ablation under Local Anesthesia*), may also play an important role in the treatment of patients with varicose disease [10].

Materials and methods

Scientific publications analyzing the description and systematization of hemodynamic disorders in varicose disease and their surgical correction methods were analyzed, with an emphasis on techniques that involve preservation of saphenous veins. The databases used for t he search were Google Scholar, PubMed, HINARI, and ResearchGate. The search terms (in English) utilized were: *" varicose veins", "chronic venous insufficiency", " hemodynamic disorders in chronic venous insufficiency", " types of hemodynamic disorders", "chronic venous disease - quality of life", "techniques of treatment for varicose veins", " preservation of saphenous veins", "CHIVA", "ASVAL".* No language limitations were set, but articles in English published after 2009 were prioritized. Included were scientific papers that addressed the principles of hemodynamic disorders in varicose disease and surgical strategies with saphenous preservation in patients with different degrees of CVI. Case reports or series of cases and original articles focusing on surgical interventions involving the removal or destruction of saphenous veins through open surgeries or endovenous treatment methods were excluded.

Results and discussions

As a result of studying the bibliographic sources, the main theories of pathological venous reflux development, types of hemodynamic disorders, and their classification principles in CVI have been identified, which would have an impact on the selection of surgical tactics for patients with varicose veins.

Theories of the development and progression of pathological venous reflux

Currently, the pathophysiology of CVI is not yet sufficiently elucidated. There are two schools that attempt to explain the development and propagation of pathological venous reflux as an "ascending" process and, respectively, a " descending" one. According to the "descending theory", CVI represents a consequence of proximal impairment, meaning it is due to an anatomical-functional failure of the proximal components of the superficial venous system: the valves at the level of the competent junction (JSF) or saphenopopliteal junction (JSP). Due to valvular incompetence at the proximal level, pathological reflux occurs in the saphenous veins, leading to increased venous pressure and stress on the distal valves. This phenomenon produces progressive distal valvular incompetence with venous dilation and formation of varicosities. The descending theory predicts the elimination of venous reflux at the primary area of incompetence, i.e., disconnecting the JSF or JSP, and is widely accepted, especially when planning the removal or endovenous obliteration of the saphenous trunk [11]. It is worth mentioning that there are numerous publications based on detailed results from Doppler Duplex scans that contest the theory of descending progression and insist on the possibility of distal evolution of CVI, whether localized or multifocal, sometimes "ascending" or "anterograde". According to these studies, the ostial valve is frequently competent (> 50% of cases), even when there is reflux in the saphenous trunk [12, 13].

According to the "ascending theory," the development of the pathological process occurs in the peripheral part of the superficial venous system of the lower limb, with cranial propagation along the saphenous trunk. This theory is based on studies that demonstrate, through Doppler Duplex evaluation, that reflux can follow a multifocal pattern, which does not always include JSF incompetence or JSP [13, 14]. A recent study, which included 1882 limbs with CVI, highlighted various types of reflux that do not always correspond to a "classic" pattern – JSF incompetence, reflux in the saphenous trunk, reflux in varicose tributaries. Based on the results obtained from t he Doppler Duplex examination, supporters of the ascending theory developed a detailed classification of the types of reflux in the superficial venous system, based on several principles – JSF competence, the presence of varicose tributaries, and the distal extension of reflux [15]:

Types of hemodynamic disorders in the venous system

great saphenous vein (VSM) (S.Chastanet, P.Pittaluga, 2013)

Type 1: dilated varicose tributaries with competent junction and VSM

Type 2: reflux in VSM, without the presence of varicose veins

Type 2a: competent junction SFJ, reflux in VSM does not extend to the malleolar level

Type 2b: competent junction SFJ, reflux in VSM extends to the malleolar level

Type 2c: incompetent SFJ, reflux in VSM does not extend to the malleolar level

Type 2d: incompetent SFJ, reflux in VSM spreads up t o the malleolar level.

Type 3: dilated varicose tributaries with reflux in VSM, but competent junction.

Type 3a: reflux in VSM does not spread up to the malleolar level.

Type 3b: reflux in VSM spreads up to the malleolar level.

Type 4: dilated varicose tributaries with reflux in VSM and incompetent junction.

Type 4a: reflux in VSM does not spread up to the malleolar level.

Type 4b: reflux in VSM extends to the level of the malleolus.

Type 5: incompetent JSF, reflux into a branch of JSF with competent VSM.

Based on the ascending theory, therapeutic recommendations for CVI are focused on redirecting venous circulation through minimally invasive interventions, emphasizing that disconnecting VSM to treat JSF i ncompetence should not be viewed as the only standard therapeutic option. In this case, treatment should begin with the removal of the 'venous reservoir' (varicose veins) while preserving the saphenous trunk. Moreover, the physiological role of the saphenous vein, as well as the hemodynamic and clinical effectiveness of these interventions, is noted to be just as good as after the 'classic' ones, along with the potential for preserving the saphenous vein as a material for possible coronary or peripheral arterial bypass in the patient's future [9, 15].

The concept of venous shuntî

Depending on the reflux source and the drainage pathways, C.Franceschi developed the concept of a venous shunt. The shunt represents a venous pathway that transports not only physiological drainage but also pathologically diverted blood (a veno-venous derivation). It begins at a reflux starting point (*escape point*) and ends at its so-called re-entry point (*re- entry point*). Since multiple reflux pathways can be created along the superficial venous system, a comprehensive specific classification has been proposed.

According to the principles developed by C.Franceschi, the lower limb veins are divided into four anatomical groups based on their position relative to the fasciae, with the abbreviation N (from *network* - rețea). The author defines the following four venous anatomical networks:

Network "N1" – all veins located in the deep compartment, bounded by the muscular fascia (in other words – deep veins and perforating veins);

The "N2" network – all the veins that are located directly on the fascia muscular and beneath the membranous layer (saphenous fascia), or the so-called interfascial veins (the trunks of the saphenous veins, Giacomini intersaphenous vein, the proximal segment of the anterior accessory saphenous vein);

Network "N3" – all veins that are located between the superficial fascia and the skin: saphenous tributaries and all tiny reticular veins, capillaries, and telangiectasias;

Network "N4" – these are N3 type veins (i.e., epifascial veins) that provide the connection between the veins in network N2. Network N4 is sub-classified into two types: longitudinal N4 – connections between segments of the same N2 vein (for example, a subcutaneous tributary that connects two points of the VSM trunk) and transverse N4 – connections between two different veins in network N2 (for instance, an epifascial tributary that connects the VSM to the small saphenous vein – VSP). Other relevant veins for describing pathological venous shunts are considered to be the suprapubic veins, marked as "P", noting the pelvic reflux source.

Based on the anatomical classification presented, under physiological conditions, the venous blood flow has the following directions:

- N3 → N1: the epifascial tributaries drain directly into the deep veins;
- N3 → N2: the epifascial tributaries drain into the tr unk of the saphenous veins;
- N2 → N1: the saphenous veins drain into the deep veins.

The direction of venous return from the extremity is from network N3 to network N2 and then to network N1. If this order is disturbed, a pathological venous flow occurs, called "reflux," with the reversal of flow direction from the deep veins to the superficial ones (N1 → N2 or N1 → N3) or from the saphenous veins to the tributaries (N2 → N3).

In this context, a venous shunt is defined as a pathological variant of blood flow with any subversion of the hierarchical order of venous drainage presented above. The overload of venous circulation within the shu nt leads to an excess of transmurals pressure and to the corresponding clinical consequences of CVI [16, 17].

According to the principles and terminology developed by C.Franceschi, shunts are classified into 6 types based on their origin, path, involved venous networks, and re-entry points (*re-entry points*). From a hemodynamic perspective, three main categories of s hunts are described based on anatomy and the characteristics of venous circulation disorders:

- Closed shunts – *closed shunts* (CS)
- Open deviated shunts – *open deviated shunts* (ODS)
- Open bypassing shunts – *open bypassing shunts* (OBS) or, according to Franceschi's original publication, – *open vicarious shunt*) [16, 17].

Category I. Closed shunts. Closed shunts create a " short-circuit" between the starting point of the reflux (usually an incompetent junction) and the re-entry point (in most cases – a perforating vein of the calf). During the relaxation of the calf muscle pump, a closed shunt becomes overloaded with venous blood flow from a deeper compartment to a more superficial one. Most cases of primary venous insufficiency, which involve

the anatomical networks N2 and N3, are determined by the circulatory disturbances characteristic of closed shunts [16, 17].

Category II. Deviated open shunts. Deviated open shunts are not associated with the phenomenon of recirculation, and the venous blood, which alters the anatomical compartment (from N2 to N3), does not return to its origin but discharges into another vein in network N2 or directly into N1 through a perforating vein *re-entry*. A deviated open shunt is active only during muscular "diastole"; during the "systole" of the calf musculature, all involved veins ensure circulation in the physiological direction, adhering to the hierarchy rules of anatomical compartments [16, 17].

Category III. Open bypass shunts. Open bypass shunts represent a natural bypass for patients with deep venous obstructions (post-thrombotic syndrome, extrinsic venous compression), thus diverting blood flow into a more superficial vein in the network to bypass the obstacle. Typically, the "exit" point is located distal to the obstruction area, while the re-entry point is near the obstacle. Clinically, this shunt serves as a compensatory mechanism and should not be surgically disconnected [16, 17]. To comprehensively systematize the various types of pathological venous reflux in patients with varicose disease and to support the reasoned selection of the optimal volume of surgical intervention, the three categories of shunts have been divided into 6 types.

Venous shunt "type 0"

The "type 0" shunt is represented by a segment of incompetent saphenous vein (of varying length), with the origin of reflux coming from a saphenous tributary and re-entering the deep venous system through a direct perforator. Regardless of the length of the incompetent saphenous vein segment, "type 0" shunts are hemodynamically benign and are not associated with the development of chronic venous insufficiency. The "type 0" shunt could be defined by the consecutiveness of the venous networks as N3 → N2 → N1. Given that there are no pathological compartmental jumps, the definition of pathological venous shunt is not actually fulfilled. However, since the definition of reflux is met (diastolic flow in the opposite direction of physiological flow), lasting more than 0.5-1 s, this situation has been defined as a "type 0" shunt (figure 1).

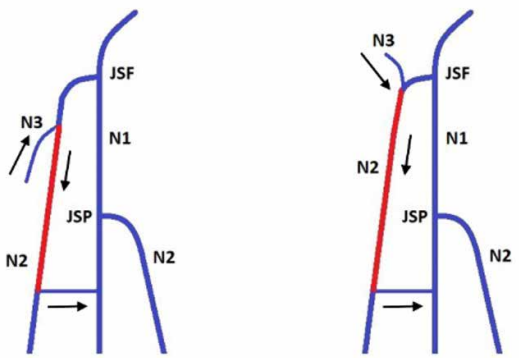

Figure 1. *It is a "type 0" shunt. The tributaries of the VSM (N3) with the physiological direction of blood flow drain into the incompetent segment of the VSM (N2) with subsequent reflux spreading distally and re-entry of blood through a perforator into the deep veins (N1).*

The amount of blood is exactly that which the VSM is capable of draining, coming from tributaries and directed towards the deep veins through a perforator. It is important to note that, depending on the length of the affected segment, a "type 0" shunt could be a precursor to a "type 2" shunt (described further).

Venous shunt "type 1"

This type of venous circulation disorders always affects the saphenous veins (N2), with the source of reflux being the deep venous system (compartmental jump N1 → N2). A "type 1" shunt occurs when the competent junction (JSF) is incompetent, the reflux extends along the saphenous trunk, and re-enters the deep system through a perforating vein (N1 → N2 → N1), (figure 2). An alternative source of reflux may be the incompetent perforating vein in the adductor femoral canal region. Thus, in this type of shunt, blood recirculates from the deep venous system a nd back, without deviation of the reflux through an incompetent tributary from the network N3.

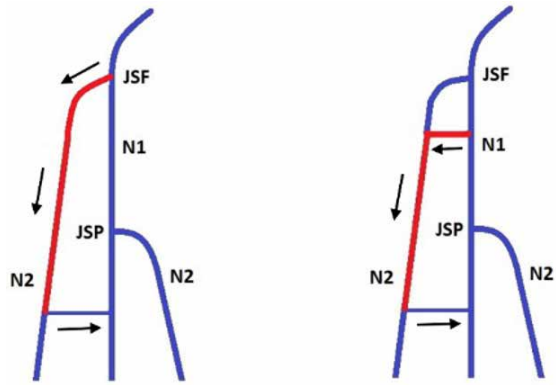

Figure 2. *It is a "type 1" shunt. Through JSF or the perforator of the adductor canal, venous blood drains from the common femoral vei n or the superficial femoral vein (N1) into the incompetent trunk of the VSM (N2), flows distally, and re-enters the deep veins (N1) through a direct perforator from the saphenous vein. No incompetent tributaries are found.*

Venous shunt "type 2"

The "type 2" shunt is characterized by a pathological compartmental jump N2 → N3. In most cases, "type 2" shunts are open deviated shunts (without recirculation), draining reflux into a

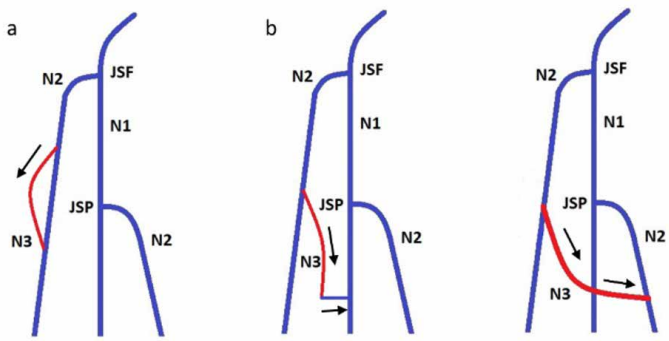

Figure 3. *Closed "type 2" shunt (a): blood from competent VSM (N2) loads a subcutaneous tributary with reflux (N3), which drains back into the same saphenous vein (N2 → N3 → N2). Open "type 2" shunt (b): blood spills from competent VSM (N2) into a tributary with reflux (N3), which subsequently drains into the VSP (N2) or into the deep veins (N1) via a perforator.*

perforating vein (figure 3b) or into the other saphenous vein (VSM → VSP, VSP → VSM). In open "type 2" shunts, the venous system of the lower limb is generally not overloaded due to the absence of recirculation; however, patients may report an aesthetic defect caused by the presence of a varicose subcutaneous vein [17].

The "type 2" shunt can also be of the closed type. In this case, the reflux source originates from a saphenous vein (N2) with normal antegrade flow. This loads a tributary (N3), which, further distally, reconnects with the saphenous trunk (figure 3a). This type of shunt is rarely encountered.

In turn, the "type 2" shunts are classified into three groups:

- **A** – no reflux is found in the saphenous vein above the incompetent tributary (figure 3a);
- **B** – no reflux starting point is found in the saphenous vein, but the saphenous trunk above the tributary with reflux is incompetent. No direct draining perforator li nking the saphenous trunk to the deep venous system can be identified (figure 4a);
- **C** – the starting point of the reflux in the saphenous vein cannot be found, and the saphenous trunk above the tributary with reflux is incompetent, with a draining perforator located along the course of the saphenous vein (figure 4b).

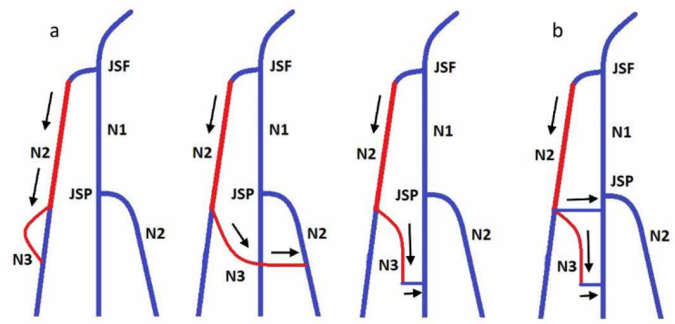

Figure 4. *Shunt "type 2B" (a): reflux in VSM has no identifiable source (escape point), but burdens an incompetent tributary that drains back into VSM (closed shunt); in another saphenous vein (open deviated shunt) or into an indirect perforating vein (open deviated shunt). Shunt "type 2C" (b): reflux in VSM without an identifiable source, draining through a direct perforating vein from the saphenous vein into the deep venous system and through a tributary (open deviated shunt).*

Venous shunt "type 1 + 2"

As mentioned above, the "type 1" shunt in its "pure" form, that is, without incompetent tributaries, is quite rare. Most often, alongside the reflux affecting the trunk of the great saphenous vein, which drains through a perforator, incompetent varicose tributaries also develop. This condition is referred to as a "type 1 + 2" shunt, as both configurations of venous circulation disorders are present.

Venous shunt "type 3"

Type 3 shunt is the most common reflux pattern encountered in clinical practice. The reflux starts at the level of incompetent competent junction, extends along the saphenous trunk, discharges into a tributary, which ultimately drains into the deep venous system through a perforator (N1 → N2 →). The entire volume of blood reflux passes into one or more epifascial tributaries (N3), representing the key moment in terms of the pathogenesis of chronic venous insufficiency. The return of blood to the deep venous system can occur i n different ways: through an indirect perforating vein (figure 5a), back into the more distal VSM trunk (figure 5b and c), or into the VSP, which can therefore also become incompetent (figure 5d).

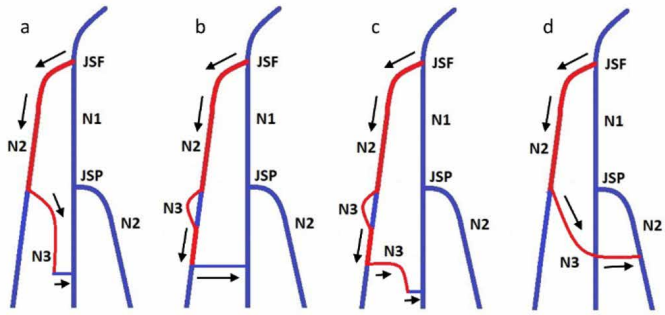

Figure 5. *Shunt "type 3". Reflux from the common femoral vein through incompetent JSF extends along the VSM and drains back into the deep venous system through a tributary. The segment of the VSM immediately distal to the refluxing tributary is competent.*

Type 4 Shunt I

In the case of the "type 4" shunt, the most common source of reflux comes from the pelvic veins or, less frequently, a perforating vein that loads the saphenous vein through a subcutaneous tributary. In this type of shunt, there is no r eflux through the incompetent junction or at the level of a direct perforating vein of the thigh. The re-entry of retrograde venous flow into the deep system occurs through a drainage perforator on the saphenous vein. In the isolated variant, this type of reflux is occasionally diagnosed, and patients ge nerally seek medical assistance when visible varicose veins appear due to the development of the "type 4+2" shunt (figure 6).

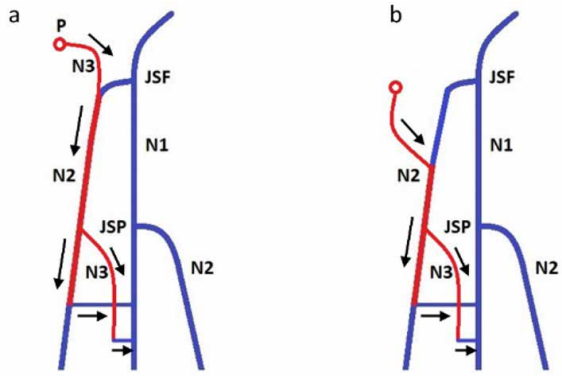

Figure 6. *Type "4+2": (a) pelvic subtype – the reflux in VSM has as its source an inguinal affluent of the saphenous arch (P → N2 – N1), (b) perforating subtype – the source of reflux in VSM is an incompetent perineal, inguinal, or gluteal perforator (N*

1 N – N – N).

→ 3 2 1

I am venous "type 5" I

In the "type 5" shunt, the reflux source is the same as in the "type 4" shunt. However, the re-entry is different. Unlike the "type 4" shunt, there is no perforator *re-entry*, but a tributary appears at the lower end of the incompetent segment of VSM, providing drainage for the reflux.

There are many possible reflux sources in the pelvic shunt types "4" and "5":

- The reflux enters the VSM through the tributaries of the saphenous arch and may begin just below the saphenofemoral confluence (between the terminal and pre-terminal valves).

- The reflux comes from the labial veins through a varicose reticular venous network on the lower thigh, overflowing into the posterior accessory saphenous vein and then into the VSM. A varicose reticular network can usually be observed on the posterior side of the thigh.

- Reflux from a tributary of the arch burdens the anterior or posterior accessory saphenous veins, which subsequently derive reflux through a tributary into one of the two saphenous trunks. The VSM is the most commonly involved trunk.

Venous shunt "type 6"

All other reflux variants without the involvement of the saphenous veins can be defined as a "type 6" shunt (figure 7), [17, 18].

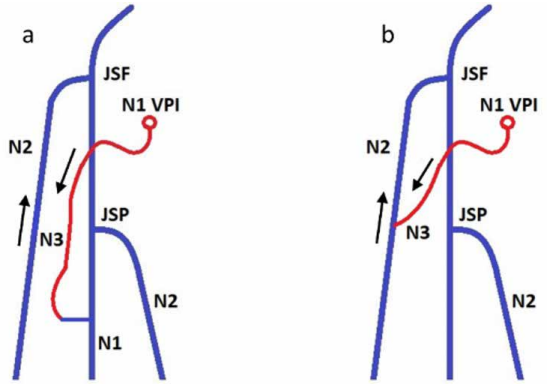

Figure 7. *Shunt "type 6". Reflux from the deep veins through an incompetent perforator into a subcutaneous tributary. The drainage path can be another perforator (a) or a competent saphenous vein (b).*

Combined shunts

According to statistics, two different types of shunts often coexist: a closed shunt with a reflux point from N1 to N2 and a re-entry point from N2 to N1, which in turn is associated with an open diverted shunt with a reflux point from N2 to N3 and a re-entry point from N3 to N1. For the selection of the therapeutic strategy, both shunts must be correctly identified during the ultrasonographic examination.

Shunt types "1" and "3" represent the most common types of hemodynamic disorders, accounting for approximately 60% of cases among patients with chronic venous disease. In these types of shunt, the saphenous veins are loaded with a retrograde blood flow wave, which originates directly from the deep veins. Shunt types " 4" and "5" are the next most common group (about 25% of cases), followed by shunt type "2" (approximately 15% of cases). Shunt "type 6" is diagnosed quite rarely [19].

The differentiation between the most common types of shunts (‘1’ and ‘3’) and the less frequent types of shunts (‘ 4’ and ‘5’) has clear therapeutic implications. This is due to the importance of the incompetent tributary or tributaries (types of shunt ‘3’ and ‘5’) as points of reflux re-entry. In the opinion of specialists who practice the surgical treatment strategy with saphenous preservation, the differentiation between a refluxing saphenous vein drained by a perforator and one drained by a tributary is crucial in selecting the volume of intervention. The differentiation is made using the Doppler Duplex examination during a special test known as the ‘reflux elimination test’ (*reflux elimination test – RET*), [16, 17]. In the patient’s orthostatic position, during the Doppler Duplex examination, digital compression of the refluxing tri butary will be performed to assess whether this alters the reflux in the VSM (figure 8). If reflux in the VSM persists, the RET is considered negative, which implies the existence of a draining perforator of the VSM. The test is positive if the reflux from the VSM disappears during digital compression of the tributary [20].

Figure 8. *Reflux elimination test (RET). Pathological venous reflux i n VSM, demonstrated with Doppler Duplex (top right). Digital compression of the incompetent tributary on the thigh (left). Disappearance of reflux in VSM after digital compression of the incompetent tributary (bottom right).*

The multitude of hemodynamic variants of venous reflux in patients with CVI explains the emergence and implementation in phlebological practice of numerous techniques to eliminate this reflux, which are not necessarily limited to the complete destruction of the main venous trunk. Depending on the clinical profile of the patient, the most suitable technique is identified, based on the advantages and disadvantages of each. Eliminating axial venous reflux to reduce venous overload is the decisive factor in the successful treatment of varicose veins. The classical method – surgical ablation of the saphenous trunk, is generally considered the gold standard. At the same time, modern surgical approaches are minimally invasive, cosmetically appealing, use local anesthesia, and are performed under ultrasonographic control. These surgical interventions can be carried out on an outpatient basis and demonstrate good long-term results, similar to or even superior to traditional open surgical methods.

Although endovenous ablative therapies are preferred by both patients and doctors due to their potential advantages, such as less pain, shorter hospital stay, faster recovery, technical simplicity, and short procedure time, open surgical techniques still play an essential role and have definitive indications in the arsenal of conte mporary surgical techniques [19, 21]. Treatment selection continues to depend on many factors, especially i ndividual anatomical characteristics and stage of the disease [10]. There are also alternative surgical techniques, which, however, are based on fundamentally different principles. CHIVA and ASVAL are two therapeutic strategies characterized by treating CVI without destroying or removing the saphenous trunk [11].

CHIVA is a minimally invasive surgical technique for the treatment of varicose veins. The purpose of CHIVA is to eliminate venous-venous shunts while preserving the saphenous vein and normal venous drainage of the soft tissues of the limb [22]. As a result, the saphenous veins of many patients can regain competence after interruption of the incompetent tributary. This can be observed after CHIVA and ASVAL [19]. Based on the results of research conducted in the context of the CHIVA method, it has been demonstrated that recurrences of CVI are less frequent if the saphenous veins are preserved during the surgical intervention. Thus, the "organ-conserving" approach should be taken into consideration in phlebology, as well as in all other surgical fields [17].

Another surgical technique, which is limited to the ablation of varicose tributaries, is ASVAL. This method ensures the preservation of the main veins of the superficial venous system, is safe, and is effective against the symptoms of chronic venous insufficiency. Potentially, the ASVAL strategy can be applied to a relatively large group of patients, with few exceptions identified through the application of preoperative hemodynamic and clinical criteria [20]. According to the results obtained by the authors of this technique, a reduction in the degree of reflux in the VSM after isolated phlebectomy of the varicose tributaries was observed, along with a significant decrease in the diameter of the VSM. These data suggest that both venous hypertension and reflux in the VSM can be significantly improved or abolished through treatment focused solely on the saphenous tributaries [15]. Therefore, the 'destructive' treatment of an incompetent saphenous vein is not always the only relevant option [23]. Additionally, further advanced study of these surgical strategies is necessary, especially through prospective studies conducted not by the authors of the mentioned methods, but in medical centers abroad – in order to obtain a clearer formulation of indications and contraindications, assess the changes in the preserved VSM during the long-term postoperative period, and to identify potential evolving complications and predictive factors for recurrence [18].

Conclusions

Venous circulation disorders in patients with CVI are characterized by an extremely wide variety and advanced complexity of anatomical configurations. The i mplementation of vascular imaging examination using Doppler Duplex in medical practice has allowed for the identification and systematization of multiple variants of pathological venous reflux and has demonstrated the necessity of utilizing personalized treatment methods. A meticulous analysis of the sources, pathways of spread, and drainage of reflux in each individual case ensures a clear understanding of the pathophysiology and is critically valuable in selecting the optimal curative approach. The application of surgical strategies for preserving the saphenous veins in medical practice must be based on the concurrent implementation of venous reflux classification systems developed by the authors of these methods.

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